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PI Name:	Held, Kathryn Ph.D.		
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PI Email:	kheld@partners.org	**Fax:**	FY 617-724-8320
PI Organization Type:	NON-PROFIT	**Phone:**	617-726-8161
Organization Name:	Massachusettes General Hospital		
PI Address 1:	Radiation Oncology		
PI Address 2:	55 Fruit Street		
PI Web Page:			
City:	Boston	**State:**	MA
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
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In space, astronauts are exposed to a complex and unique radiation environment composed of trapped proton and electron radiation, galactic cosmic rays (GCR; energetic protons and heavy charged ions, HZE), and solar particle events (SPEs), primarily protons. Although these radiations can result in appreciable cumulative doses to astronauts during long-duration missions, the exposures occur at low fluences and fluence rates, such that particle traversals through individual cells in an astronaut's body may be well separated in tissue location and time. Under such conditions, so-called "non-targeted" effects such as bystander responses may have increased significance. The radiation-induced bystander effect is the occurrence of biological responses in unirradiated cells near to or sharing medium with cells actually traversed by radiation. We hypothesize that at the low fluences of protons and HZE particles to which astronauts are exposed in space, a significant fraction of the cellular changes may be due to bystander responses, in addition to the damages caused in cells directly traversed by ionizing particles or their associated delta rays. We have the

Task Description:	following specific aims. 1. Test the hypothesis that bystander effects occur at particle fluences so low that only 1-2 % of cells in a culture dish are traversed by an energetic proton or HZE particle. 2. Extend our low fluence work to studies using repeated exposures of cells to low fluences and "mixed" (actually sequential) beams of more than one particle type. This will test the hypothesis that cells that have survived an initial traversal and sent a bystander signal, can again send a signal when traversed once again by a particle or can subsequently respond as bystanders to a "hit" in a neighboring cell. 3. Our studies have indicated that reactive oxygen species (ROS) are at least partially involved in transmittal of the Fe ion-induced bystander signal from "hit" to unirradiated cells. We will test the hypothesis that non-nuclear damage-initiated, metabolically produced ROS in "hit" cells are involved in the early steps of bystander signaling. Additionally, we will investigate chemical countermeasures to reduce the bystander responses based on removal of ROS or prevention of their formation. 4. In tissues, cells do not exist in isolation, and there is increasing interest in how signaling between cell types (especially stromal and epithelial) influences cellular responses to stresses. We will test the hypothesis that radiation-induced bystander signals can be sent between differing cell types, but the characteristics of the responses depend on the cell types and the types of radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although our studies are specifically geared to providing information on the possible biological effects, especially cancer induction, of low fluences of energetic particles on astronauts, the data may have relevance to cancer therapy and the effects of low doses of radiation on radiation workers (e.g., those involved in clean-up of radioactive sites) and the general public. With regards to cancer, there is increasing use of protons (energies of 65-250 MeV) as an effective treatment modality in radiation oncology because they provide a great therapeutic advantage in some types of cancers due to exquisite dose localization. Over 40,000 patients have been treated world-wide with protons, and several new proton centers have opened in the last few years or are under construction in the US. However, despite the clinical success of protons, some concerns exist regarding the possible side-effects, including the possibility of cancer caused by the low doses received by normal tissues. Additionally, there is increasing interest in the use of carbon ions in cancer therapy, with carbon ion treatment centers now in use in Japan and Germany, and the potential being discussed in the US. Our studies could provide important and useful new insight on the effects of these energetic particles on biological systems and the mechanisms by which they act. Such findings would be expected to translate into helping to develop effective ways to reduce possible side-effects, including cancer, from the treatments.
Task Progress:	Previously we reported the establishment in our lab of a transwell insert culture method which we have used to demonstrate media-mediated bystander responses, using a number of different endpoints, in AG01522 normal human fibroblasts after low LET X-irradiation as well as 1 GeV/n iron ions. We have now obtained similar results with 1 GeV protons. Using micronuclei (MN) formation, a measure of DNA damage, as an endpoint in irradiated and bystander cells, we have found that MN increase with dose in irradiated cells, with Fe ions being significantly more efficient at producing MN than protons (e.g., about 2.5-fold more MN at 0.5 Gy with Fe than with protons). However, in the unirradiated bystander cells sharing medium with the irradiated cells, MN formation increases about 2-fold at the lowest radiation dose of iron or protons used in this study (10 cGy), then remains constant to the highest dose and irrespective of whether iron ions or protons were used. Similar dose response data for irradiated and bystander cells have been obtained using the endpoints of formation of foci of DNA repair-related proteins, gH2AX or 53BP1, as markers of DNA double strand breaks, using doses of 5 cGy and above. Although at doses above about 10 cGy the magnitude of the bystander effects is small compared to the damage from direct irradiation, at lower doses the bystander responses make a significant contribution to the overall cell responses. In the period of this report, our work at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory has concentrated on extending these studies to very low fluences of high energy iron ions and protons (aim 1). The data show that media-mediated bystander effects are seen after 1 GeV/n iron ion exposures as low as 0.02 particles/cell (< 2% of cells traversed by an ion; dose about 0.45 mGy) for the endpoint of MN and down to 0.008 particles/cells for 53BP1 foci. We are currently investigating lower doses with the MN endpoint. In contrast, with 1 GeV protons, bystander responses are not seen below fluences of 0.2 or 2 protons/cell with the MN and 53BP1 foci endpoints, respectively, but at those fluences or higher there is a 2-fold increase in damage in bystander cells. In directly irradiated cells, much higher fluences (equivalent to 5 cGy) are needed to produce a measurable increase in MN with both ions. Interestingly, with the 53BP1 foci endpoint, the fluence-dependence data for media-mediated bystander responses are similar to the production of foci in the irradiated population, suggesting the response in the irradiated population is dominated by bystander effects. At this time we do not understand the different responses between the endpoints of MN and 53BP1 foci. The results just presented were obtained using exponentially growing cells. Much of the pioneering work by others in bystander responses has been done using confluent cells, and most cells in the human body are not exponentially growing, so we have begun to investigate whether results similar to those just described are obtained with confluent cells. This is also important from a mechanistic standpoint, since it has been proposed that gap junctions are important for the bystander effect in confluent cells, but gap junctions are not relevant in studies done using our insert system. Our preliminary data show that when confluent AG01522 cells are exposed to low fluences of Fe ions or protons, no bystander effects (MN formation and 53BP1 foci induction) are observed in the unirradiated cells sharing medium with irradiated confluent cells regardless of whether the bystander cells are confluent or in exponential growth phase. The findings agree with results we have obtained recently using X-rays. Moreover, with X-rays, bystander 53BP1 foci formation was observed in confluent recipient cells when the irradiated cells were in exponential growth phase. This suggests that while exponential AG01522 cells send medium-mediated bystander signals, confluent AG01522 cells do not send medium-mediated bystander signals, but both exponential and confluent AG01522 cells respond to medium-borne signals. Starting during the funding period of our previous NASA grant and continuing with this new grant, we have been among the pioneering groups at NSRL to use the "mixed", actually sequential, beams of Fe ions and protons. Our initial data suggested that, depending on the time between irradiations, bystander cells can "adapt" to the second signal, i.e., a

	smaller bystander response was seen when irradiated cells were “hit” twice with time (3-24 h) between the hits, than when the same dose was given in a single irradiation. These initial studies were conducted using 10 cGy of each beam, a relatively high dose compared to the region of greatest interest, so in our recent runs at NSRL we have begun to assess the signaling at the very low particle fluences more realistic of the space environment (iron fluences down to 2000 particles/cm2 and proton doses of 1 cGy), as proposed in aim 2 and are obtaining data on times between 0 and 3 h separation of the beams. In recently published work we showed that SOD, catalase, and c-PTIO, which remove superoxide, hydrogen peroxide and nitric oxide, respectively, when added to medium reduce the Fe ion-induced bystander effects, indicating a role of reactive oxygen species (ROS) in ion-induced bystander effects. We have now extended that work to study the effect of DPI, an inhibitor of the membrane-localized superoxide-generating enzyme NADPH oxidase. Our findings indicate that DPI reduces bystander effects after both iron ion and proton irradiation, but has no effect on irradiated cells, suggesting the source of the superoxide involved in bystander signaling may, at least in part, be NADPH oxidase. Second, we have been studying the impact of adding low, non-toxic concentrations of hydrogen peroxide to cells before or after ion irradiation, to assess whether chronic oxidative stress alters cellular response to irradiation or the bystander effect. To date, our data show that if bystander AG0 cells are exposed to low levels of hydrogen peroxide before co-culturing with irradiated cells, there is no radiation-induced bystander effect in terms of MN formation, but if bystander cells are put under low-level oxidative stress after the start of co-culturing with irradiated cells, the bystander MN formation is not changed.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
Abstracts for Journals and Proceedings	Yang H, Purschke M, Spantchak Y, Held KD. "Bystander Effects Induced By Low Fluences of Energetic Protons and Iron Ions in Human Normal Fibroblasts." Oral presentation at 18th Annual NASA Space Radiation Investigators’ Workshop, Sonoma, CA, July 13-15, 2007. Proceedings of the 18th Annual NASA Space Radiation Investigators’ Workshop, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Yang H, Held KD. "Responses in Irradiated and Bystander Cells Exposed to Very Low Fluences of Iron Ions or Energetic Protons." Poster presentation at the NASA Human Research Program Investigators’ Workshop, League City, TX, February 4-6, 2008. Proceedings of the NASA Human Research Program Investigators’ Workshop, League City, TX, February 2008. , Feb-2008